



Safety and efficacy of astegolimab for COPD with frequent exacerbations regardless of baseline blood eosinophil counts (ALIENTO and ARNASA): randomised, double-blind, placebo-controlled, phase 2b and 3 trials

Alberto Papi*, Neil J Greening*, Surya P Bhatt, Nicolas Roche, Bartolome Celli, Jadwiga A Wedzicha, Àlvar Agustí, Ruth Tal-Singer, MeiLan K Han, Wim Janssens, Nicola A Hanania, Parameswaran Nair, Peter Bremner, Konstantinos Porpodis, Yoko Shibata, Stephanie Korn, Ting Yang, Oliver Gordon, Rebecca Saenz, Julie Ng, Dorothy S Cheung, Jacob Devine, Michele A Grimbaldston, Wenhui Zhang, Xiaoying Yang, Divya Mohan, Claus F Vogelmeier†, Christopher E Brightling†, on behalf of the ALIENTO and ARNASA investigators‡

Summary

Background Interleukin-33 and its receptor, ST2, are implicated in neutrophilic and eosinophilic inflammation during chronic obstructive pulmonary disease (COPD) exacerbations. We aimed to assess the efficacy and safety of astegolimab, an anti-ST2 human IgG2 monoclonal antibody, which were evaluated in two COPD pivotal trials.

Methods In two randomised, double-blind, placebo-controlled trials (phase 2b ALIENTO and phase 3 ARNASA), current or former smokers with COPD and a history of frequent exacerbations, irrespective of baseline blood eosinophils, were randomly assigned (1:1:1; stratification by smoking status and region) to receive subcutaneous astegolimab 476 mg every 2 weeks, every 4 weeks, or placebo, alongside optimised inhaled maintenance therapy over 52 weeks. The primary endpoint (analysed in participants receiving one or more doses) was annualised rate of moderate or severe exacerbations. Missing data were considered similar to data from other participants in the same treatment group with the same baseline characteristics. The trials were registered with ClinicalTrials.gov (NCT05037929 and NCT05595642).

Findings In ALIENTO, 1301 participants (astegolimab every 2 weeks, n=433; astegolimab every 4 weeks, n=437; and placebo, n=431) initiated treatment between Oct 5, 2021, and Feb 19, 2024. In ARNASA, 1375 participants (astegolimab every 2 weeks, n=459; astegolimab every 4 weeks, n=459; and placebo, n=457) initiated treatment between Jan 9, 2023, and June 25, 2024. Adjusted rate ratios versus placebo for the primary endpoint were 0·85 (95% CI 0·72–1·00; p=0·049) for astegolimab every 2 weeks and 0·93 (0·79–1·10; p=0·38) for astegolimab every 4 weeks in ALIENTO, and 0·85 (0·72–1·01; p=0·068) for astegolimab every 2 weeks and 0·82 (0·70–0·98; p=0·024) for astegolimab every 4 weeks in ARNASA. Adverse events were balanced between treatments, with most participants experiencing one or more adverse events [1093 [84·0%] of 1301 participants in ALIENTO and 1176 [85·5%] of 1375 in ARNASA). The most common non-COPD adverse event was nasopharyngitis in ALIENTO and upper respiratory chest infection in ARNASA. Deaths occurred in 40 (3·1%) of 1301 participants in ALIENTO and in 44 (3·2%) of 1375 participants in ARNASA, and were balanced across treatment groups. Across both trials, a total of three deaths (0·1%) were considered to be related to treatment by investigators.

Interpretation In ALIENTO, astegolimab every 2 weeks was associated with a lower annual rate of exacerbations versus placebo in patients with COPD and a history of frequent exacerbations. In ARNASA, these findings did not meet statistical significance. Together, these findings suggest a role for targeting the ST2/IL-33 pathway to reduce the frequency of COPD exacerbations in patients with limited treatment options.

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Introduction

Chronic obstructive pulmonary disease (COPD) is defined as a heterogeneous lung condition associated with chronic respiratory symptoms, persistent and often progressive airflow obstruction that is incompletely reversible, and airway inflammation.^{1,2} Pharmacological treatment has historically been based

on inhaled bronchodilators (long-acting beta agonists [LABA], and long-acting muscarinic antagonists [LAMA]), with or without inhaled corticosteroids (ICS), and is primarily guided by symptoms and exacerbation history.¹ However, many patients still have exacerbations while receiving optimised maintenance therapy.^{3,4} Frequent or severe exacerbations increase risk of future

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*Joint first authors

†Joint senior authors

‡Members listed in appendix 1 (pp 5–12)

Pulmonary Division, University of Ferrara, St Anna University Hospital, Ferrara, Italy (Prof A Papi MD); Institute for Lung Health, Department of Respiratory Sciences, University of Leicester, Leicester, UK (N J Greening PhD, C E Brightling FMedSci); Division of Pulmonary, Allergy, and Critical Care Medicine, University of Alabama at Birmingham, Birmingham, AL, USA (Prof S P Bhatt MD); Université Paris Cité, APHP Centre, Hôpital et Institut Cochin, Paris, France (Prof N Roche MD); Mass General Brigham Hospital, Harvard University Medical School, Boston, MA, USA (Prof B Celli MD); National Heart and Lung Institute, Imperial College London, London, UK (Prof J A Wedzicha FMedSci); University of Barcelona, Barcelona, Spain (Prof À Agustí MD); Respiratory Institute, Hospital Clinic, Barcelona, Spain (Prof À Agustí); FCRB-IDIBAPS, Barcelona, Spain (Prof À Agustí); CIBER Enfermedades Respiratorias, ISCIII, Spain (Prof À Agustí); Global Allergy & Airways Patient Platform, Vienna, Austria (R Tal-Singer PhD); University of Michigan, Ann Arbor, MI, USA (Prof M K Han MD); Department of Chronic Diseases and

Metabolism (CHROMETA), University Hospital Leuven, KU Leuven, Belgium (Prof W Janssens MD); Section of Pulmonary and Critical Care Medicine, Baylor College of Medicine, Houston, TX, USA (Prof N A Hanania MD); Department of Medicine, McMaster University, Hamilton, ON, Canada (Prof P Nair MD); Research Institute of St Joe's Hamilton, St Joseph's Healthcare Hamilton, Hamilton, ON, Canada (Prof P Nair); University of Notre Dame, Fremantle, WA, Australia (Prof P Bremner MD); Pulmonology Department, G Papanikolaou Hospital, School of Medicine, Aristotle University of Thessaloniki, Thessaloniki, Greece (K Porpodis MD); Fukushima Medical University, Fukushima, Japan (Prof Y Shibata MD); IKF Pneumologie Mainz, Mainz, Germany (S Korn MD); Thoraxklinik Heidelberg, Heidelberg, Germany (S Korn); China-Japan Friendship Hospital, Beijing, China (T Yang MD); Roche Products, Welwyn Garden City, Hertfordshire, UK (O Gordon PhD); Genentech, South San Francisco, CA, USA (R Saenz MD, J Ng MD, D S Cheung MD, J Devine MA, M A Grimbaldston PhD, W Zhang PhD, X Yang PhD, D Mohan MD); Department of Medicine, Pulmonary, Critical Care and Sleep Medicine, Philipps University of Marburg, German Center for Lung Research (DZL), Marburg, Germany (Prof C F Vogelmeier MD)

Correspondence to: Prof Alberto Papi, Pulmonary Division, University of Ferrara, St Anna University Hospital, 44124 Ferrara, Italy ppa@unife.it

See Online for appendix 1

Research in context

Evidence before this study

Astegolimab is a human anti-ST2 IgG2 monoclonal antibody that blocks ST2-mediated IL-33 activity. To identify the available clinical evidence on astegolimab before the current studies, we searched PubMed from database inception to May 20, 2025, using the search term “astegolimab”, and its alternative terms “RG-6149”, “RG6149”, “MSTT-1041A”, “MSTT1041A”, “RO-7187807”, “RO7187807”, and “AMG-282”. No additional search parameters were applied. We identified four previous clinical trials studying the ability of astegolimab to target the ST2/IL-33 axis in asthma, chronic obstructive pulmonary disease (COPD), atopic dermatitis, and COVID-19 pneumonia. In the first reported clinical trial of astegolimab in COPD (phase 2a COPD-ST2OP), treatment with astegolimab resulted in a non-significant 22% relative risk reduction in exacerbation rate over 48 weeks versus placebo (rate ratio 0.78 [95% CI 0.53–1.14]; $p=0.19$).

Added value of this study

The phase 2b ALIENTO and phase 3 ARNASA trials are the first pivotal, late-stage, multicentre trials to report data on targeting ST2/IL-33 in COPD in participants with a history of frequent exacerbations while receiving optimised dual or triple inhaled maintenance therapy, regardless of blood eosinophil count or

whether they are current or former smokers. In contrast to previous studies of COPD biologics that restricted eligibility based on blood eosinophil count, the ALIENTO and ARNASA trials enrolled a clinically heterogeneous population. This strengthens the generalisability of our findings to a wider COPD population, suggesting that astegolimab could provide additional control of COPD exacerbations in patients who continue to experience COPD exacerbations on optimised inhaled therapy and who are not eligible for other biological therapies. The studies also enable exploration across a wide range of clinical and biological characteristics, offering initial insight into which patient subgroups might benefit most from astegolimab.

Implications of all the available evidence

Our findings suggest a role for targeting the ST2/IL-33 pathway in COPD to reduce the frequency of moderate or severe exacerbations in patients who have scarce treatment options beyond standard inhaled maintenance therapies. Additional research could help to establish the benefit of astegolimab in patients with COPD, particularly in those with frequent moderate-to-severe exacerbations (ie, more than two per year) and with moderate airflow obstruction.

exacerbations, cardiovascular events, hospitalisation, and mortality.^{3–7}

Targeting inflammatory pathways with biologics can modulate COPD-related inflammation and improve outcomes. The first add-on biologics for inadequately controlled COPD are approved for patients with an eosinophilic phenotype.^{8–10} However, most people with COPD have predominantly neutrophilic (type 1 or type 3, or both) inflammation, whereas only a minority have eosinophilic (type 2) inflammation or a combination of both.^{11–13} Consequently, there remains a substantial unmet need for effective treatments for people with frequent COPD exacerbations despite optimised maintenance therapy, particularly for those without a type 2 eosinophilic phenotype. Additionally, baseline systemic markers of inflammation (types 1, 2, and 3) do not consistently correspond to the type of future COPD exacerbations that a patient might have. For example, patients characterised as having type 2 inflammation (based on single or multiple assessments) have been shown to have both type 2 and non-type 2 exacerbations.¹³ The biological heterogeneity that underlies COPD and its exacerbations provides a rationale for why a broad anti-inflammatory treatment option might be appealing in theory, as it could have the potential to prevent different types of exacerbations and address the unmet needs of a broad COPD population.

COPD exacerbations can have multiple underlying triggers, including pathogens, allergens, or noxious particles such as cigarette smoke.¹⁴ The response to these agents includes lung epithelial release of alarmin

interleukin-33 (IL-33); IL-33 levels are elevated in patients with COPD versus those without COPD,¹⁵ and patients with elevated plasma IL-33 levels have been shown to have more COPD exacerbations versus those with lower plasma IL-33 levels.¹⁶ The binding of IL-33 to the receptor ST2 (interleukin-1 receptor-like 1)—which is present on a wide range of immune cells including neutrophils, eosinophils, macrophages, basophils, and mast cells—activates downstream type 1, type 2, and type 3 signalling cascades that result in both neutrophilic and eosinophilic inflammation.^{17,18} Astegolimab is a human anti-ST2 IgG2 monoclonal antibody that blocks ST2-mediated IL-33 activity and subsequent inflammatory responses.^{19,20} In the phase 2a COPD-ST2OP trial ($n=81$), treatment with astegolimab resulted in a non-statistically significant 22% relative risk reduction in exacerbation rate over 48 weeks versus placebo in participants with moderate-to-very-severe airflow obstruction and two or more exacerbations within the previous year.¹⁹ Here we aimed to assess the primary efficacy and safety results from the phase 2b ALIENTO and phase 3 ARNASA astegolimab trials, the first late-stage trials targeting the ST2/IL-33 pathway in COPD, in participants with a history of frequent exacerbations regardless of their baseline blood eosinophil counts or smoking status.

Methods

Study design

The ALIENTO and ARNASA trials were pivotal, randomised, double-blind, placebo-controlled trials that

enrolled participants at 191 sites across 24 countries in ALIENTO and 319 sites across 35 countries in ARNASA (appendix 1 p 13). ALIENTO was intended to be, and designed as, a potentially pivotal phase 2b trial, with an interim futility analysis that ungated continued development of the phase 3 ARNASA trial.

The trials complied with the International Council for Harmonisation guideline for Good Clinical Practice and the principles of the Declaration of Helsinki, and were approved by a central review board or a site-specific review board in each participating country and at each participating global site (Advarra approval number for US Central Institutional Review Board: ALIENTO [Pro00054357]; ARNASA [Pro00066877]). The protocols and statistical analysis plans are available in appendix 2, including a description of changes following trial commencement.

Patient group representatives were involved in the design of both ALIENTO (COPD Foundation) and ARNASA (patient focus group and Global Allergy and Airways Patient Platform representatives). The ALIENTO and ARNASA trials were registered with ClinicalTrials.gov (NCT05037929 and NCT05595642).

Participants

Participants were aged 40–90 years in ALIENTO and 40–80 years in ARNASA, had a documented COPD diagnosis 12 months or more before screening, a history of frequent exacerbations (two or more moderate or severe exacerbations within a 12-month period in the 24 months before screening²¹ in ALIENTO [appendix 1 p 13], or within the 12 months before screening in ARNASA), and were receiving optimised dual or triple inhaled maintenance therapy per treatment guidelines at trial initiation (ICS+LABA, LAMA+LABA, or ICS+LAMA+LABA; appendix 1 p 15). Participants had a post-bronchodilator forced expiratory volume in 1 s (FEV₁)/forced vital capacity (FVC) ratio of <0.70, post-bronchodilator percentage predicted FEV₁ (%FEV₁) of ≥20% to <80%, and a modified Medical Research Council dyspnoea score of ≥2 at screening (during screening or on day 1 for %FEV₁, FEV₁, and FVC measurements in ARNASA). Participants were current or former smokers with a smoking history of ≥10 pack-years. Exclusion criteria for both trials included a current documented diagnosis of asthma or any clinically significant pulmonary disease other than COPD. There were no restrictions on blood eosinophil count or the presence or absence of chronic bronchitis (defined as patient-reported cough or sputum on most days). Demographic information related to sex, race, and ethnicity was self-reported by participants in both trials. Full eligibility criteria are described in appendix 1 (pp 19–25). Written informed consent was obtained from participants.

Randomisation and masking

A permuted-block randomisation method was used to ensure balance between treatment groups, with

stratification by smoking status and region. Treatment assignment was obtained from an interactive voice or web-based response system. Participants, site personnel, and the sponsor were masked to treatment assignment and post-baseline blood eosinophil counts during the trial, with the exception of individuals who required access to participant treatment assignments to fulfil their roles (eg, the group responsible for unmasking, clinical supply chain managers, sample handling staff, operational assay group personnel, interactive voice or web-based response system service provider, and independent data monitoring committee members).

Procedures

Following a 1–4 week screening period, participants were randomly assigned (1:1:1) to 476 mg of subcutaneous astegolimab every 2 weeks or every 4 weeks or placebo during the 52-week treatment period (appendix 1 pp 30–31). Astegolimab and placebo were supplied as sterile liquids in identical pre-filled syringes with a needle safety device and were administered by trained medical personnel. Participants received subcutaneous astegolimab or matching placebo every 2 weeks from day 1 up to week 50, with the astegolimab every 4 weeks group alternating astegolimab and placebo every 2 weeks to receive astegolimab every 4 weeks (with astegolimab as the first injection). Participants returned to the clinic every 2 weeks for assessments up to the treatment completion visit at week 52 (or the end of the additional treatment period, if applicable; appendix 1 pp 30–31), and all participants underwent a safety follow-up visit 12 weeks after their final dose of trial drug. Assessments for exacerbation endpoints were performed at every visit, and assessments for St George's Respiratory Questionnaire for COPD (SGRQ-C) and post-bronchodilator FEV₁ were done at weeks 0, 12, 24, 36, and 52. The Evaluating Respiratory Symptoms in COPD (E-RS:COPD) was assessed based on an electronic diary that was completed daily for ALIENTO and for 7 days before weeks 0, 12, 24, 36, and 52 for ARNASA. Adverse event information was collected at every visit. For further details on screening failures, trial procedures, and important protocol deviations, including major protocol deviations that occurred in ≥1% of the total population in ALIENTO and ARNASA, see appendix 1 (pp 13–15, 27–29).

Outcomes

We report the primary analysis outcomes. The primary endpoint in both trials was the annualised rate of moderate or severe exacerbations over the 52-week treatment period. Moderate exacerbations were defined as new or increased COPD symptoms that led to treatment for 3 or more days with systemic corticosteroids (>10 mg per day prednisolone or equivalent; appendix 1 p 16) or antibiotics, or both; severe exacerbations were defined as new or increased COPD symptoms that led to

See Online for appendix 2

	ALIENTO				ARNASA			
	Astegolimab every 2 weeks (n=433)	Astegolimab every 4 weeks (n=437)	Placebo (n=431)	Total (N=1301)	Astegolimab every 2 weeks (n=459)	Astegolimab every 4 weeks (n=459)	Placebo (n=457)	Total (N=1375)
Age, years								
Mean (SD)	66.8 (8.3)	67.6 (7.9)	67.5 (7.6)	67.3 (8.0)	66.8 (7.6)	66.6 (7.4)	66.8 (6.9)	66.7 (7.3)
Median (range)	67.0 (41.0–88.0)	68.0 (45.0–88.0)	67.0 (44.0–90.0)	67.0 (41.0–90.0)	68.0 (42.0–81.0)	67.0 (40.0–80.0)	67.0 (40.0–80.0)	67.0 (40.0–81.0)
≥65 years	262 (60.5%)	288 (65.9%)	288 (66.8%)	838 (64.4%)	305 (66.4%)	296 (64.5%)	299 (65.4%)	900 (65.5%)
Sex								
Male	250 (57.7%)	243 (55.6%)	234 (54.3%)	727 (55.9%)	288 (62.7%)	296 (64.5%)	302 (66.1%)	886 (64.4%)
Female	183 (42.3%)	194 (44.4%)	197 (45.7%)	574 (44.1%)	171 (37.3%)	163 (35.5%)	155 (33.9%)	489 (35.6%)
Race								
American Indian or Alaska Native	5 (1.2%)	1 (0.2%)	8 (1.9%)	14 (1.1%)	4 (0.9%)	4 (0.9%)	3 (0.7%)	11 (0.8%)
Asian	6 (1.4%)	10 (2.3%)	9 (2.1%)	25 (1.9%)	106 (23.1%)	101 (22.0%)	114 (24.9%)	321 (23.3%)
Black or African American	9 (2.1%)	12 (2.7%)	14 (3.2%)	35 (2.7%)	20 (4.4%)	34 (7.4%)	21 (4.6%)	75 (5.5%)
Native Hawaiian or other Pacific Islander	1 (0.2%)	2 (0.5%)	1 (0.2%)	4 (0.3%)	4 (0.9%)	0	1 (0.2%)	5 (0.4%)
White	397 (91.7%)	396 (90.6%)	389 (90.3%)	1182 (90.9%)	319 (69.5%)	309 (67.3%)	310 (67.8%)	938 (68.2%)
Multiple races reported	7 (1.6%)	8 (1.8%)	8 (1.9%)	23 (1.8%)	1 (0.2%)	0	2 (0.4%)	3 (0.2%)
Race not reported or unknown	8 (1.8%)	8 (1.8%)	2 (0.5%)	18 (1.4%)	5 (1.1%)	11 (2.4%)	6 (1.3%)	22 (1.6%)
Ethnicity								
Hispanic or Latino	82 (18.9%)	78 (17.8%)	83 (19.3%)	243 (18.7%)	65 (14.2%)	66 (14.4%)	71 (15.5%)	202 (14.7%)
Not Hispanic or Latino	342 (79.0%)	352 (80.5%)	334 (77.5%)	1028 (79.0%)	384 (83.7%)	377 (82.1%)	369 (80.7%)	1130 (82.2%)
Not reported or unknown	9 (2.1%)	7 (1.6%)	14 (3.2%)	30 (2.3%)	10 (2.2%)	16 (3.5%)	17 (3.7%)	43 (3.1%)
Smoking status								
Former smokers	284 (65.6%)	285 (65.2%)	281 (65.2%)	850 (65.3%)	306 (66.7%)	306 (66.7%)	304 (66.5%)	916 (66.6%)
Current smokers	149 (34.4%)	152 (34.8%)	150 (34.8%)	451 (34.7%)	153 (33.3%)	153 (33.3%)	153 (33.5%)	459 (33.4%)
Number of moderate or severe COPD exacerbations*								
Two	296/432 (68.5%)	298/437 (68.2%)	296/431 (68.7%)	890/1300 (68.5%)	316/459 (68.8%)	319/458 (69.7%)	353/455 (77.6%)	988/1372 (72.0%)
More than two	136/432 (31.5%)	139/437 (31.8%)	135/431 (31.3%)	410/1300 (31.5%)	143/459 (31.2%)	139/458 (30.3%)	102/455 (22.4%)	384/1372 (28.0%)
Blood eosinophil count								
n	433	437	431	1301	455	458	457	1370
Mean (SD), per μL	190.0 (181.2)	175.4 (111.0)	177.2 (119.6)	180.9 (140.8)	177.3 (134.6)	199.2 (255.2)	176.0 (113.1)	184.2 (179.2)
Median (range), per μL	140 (0–1900)	150 (20–720)	150 (0–860)	150 (0–1900)	140 (0–1050)	150 (0–4130)	150 (0–810)	150 (0–4130)
IQR	100–230	100–220	100–210	100–220	90–230	100–230	100–220	100–230
<150 cells/ μL	228 (52.7%)	212 (48.5%)	206 (47.8%)	646 (49.7%)	231 (50.8%)	222 (48.5%)	221 (48.4%)	674 (49.2%)
≥150 to <300 cells/ μL	143 (33.0%)	174 (39.8%)	180 (41.8%)	497 (38.2%)	163 (35.8%)	170 (37.1%)	180 (39.4%)	513 (37.4%)
≥300 cells/ μL	62 (14.3%)	51 (11.7%)	45 (10.4%)	158 (12.1%)	61 (13.4%)	66 (14.4%)	56 (12.3%)	183 (13.4%)
Mean FEV ₁ (SD), L	1.3 (0.5); n=432	1.2 (0.5); n=437	1.2 (0.5); n=430	1.2 (0.5); n=1299	1.2 (0.5); n=457	1.2 (0.4); n=458	1.3 (0.5); n=454	1.2 (0.5); n=1369
Mean %FEV ₁ (SD)	46.3 (15.4); n=432	44.6 (15.4); n=437	43.6 (15.2); n=430	44.9 (15.4); n=1299	45.2 (15.0); n=457	44.9 (14.9); n=458	46.7 (15.9); n=454	45.6 (15.3); n=1369
Mean SGRQ-C total score (SD)	60.2 (16.6); n=431	57.6 (18.1); n=433	59.5 (17.6); n=429	59.1 (17.5); n=1293	59.7 (18.6); n=455	61.7 (17.7); n=459	59.7 (18.8); n=452	60.4 (18.4); n=1366
Mean E-RS-COPD total score (SD)	16.0 (6.5); n=419	15.9 (6.7); n=420	16.0 (6.8); n=416	16.0 (6.7); n=1255	16.1 (6.5); n=453	16.7 (7.1); n=451	15.9 (6.4); n=449	16.2 (6.7); n=1353

(Table 1 continued on next page)

	ALIENTO				ARNASA			
	Astegolimab every 2 weeks (n=433)	Astegolimab every 4 weeks (n=437)	Placebo (n=431)	Total (N=1301)	Astegolimab every 2 weeks (n=459)	Astegolimab every 4 weeks (n=459)	Placebo (n=457)	Total (N=1375)
(Continued from previous page)								
Standard-of-care maintenance therapy								
n	432	436	430	1298	459	459	457	1375
ICS+LAMA+LABA	338 (78.2%)	346 (79.4%)	346 (80.5%)	1030 (79.4%)	357 (77.8%)	340 (74.1%)	326 (71.3%)	1023 (74.4%)
ICS+LABA	62 (14.4%)	52 (11.9%)	54 (12.6%)	168 (12.9%)	66 (14.4%)	78 (17.0%)	79 (17.3%)	223 (16.2%)
LAMA+LABA	32 (7.4%)	38 (8.7%)	28 (6.5%)	98 (7.6%)	35 (7.6%)	41 (8.9%)	50 (10.9%)	126 (9.2%)
ICS	0	0	0	0	1 (0.2%)	0	2 (0.4%)	3 (0.2%)
ICS+LAMA	0	0	1 (0.2%)	1 (0.1%)	0	0	0	0
LAMA	0	0	1 (0.2%)	1 (0.1%)	0	0	0	0
Use of ICS	400 (92.4%)	398 (91.1%)	401 (93.0%)	1199 (92.2%)	424 (92.4%)	418 (91.1%)	407 (89.1%)	1249 (90.8%)
Use of azithromycin	9 (2.1%)	19 (4.3%)	18 (4.2%)	46 (3.5%)	6 (1.3%)	16 (3.5%)	20 (4.4%)	42 (3.1%)
Use of roflumilast	13 (3.0%)	2 (0.5%)	15 (3.5%)	30 (2.3%)	22 (4.8%)	15 (3.3%)	11 (2.4%)	48 (3.5%)
Diagnosis of chronic bronchitis†	232 (53.6%)	223 (51.0%)	230 (53.4%)	685 (52.7%)	188 (41.0%)	213 (46.4%)	197 (43.1%)	598 (43.5%)
Data are n (%) or n/N (%), unless otherwise specified. %FEV ₁ =percentage of predicted forced expiratory volume in 1 s. COPD=chronic obstructive pulmonary disease. E-RS:COPD=Evaluating Respiratory Symptoms in COPD. FEV ₁ =forced expiratory volume in 1 s. ICS=inhaled corticosteroid. LABA=long-acting beta agonist. LAMA=long-acting muscarinic antagonist. mITT=modified intention-to-treat. SGRQ-C=St George's Respiratory Questionnaire for COPD. *Number of moderate or severe COPD exacerbations during a 12-month period within the 24 months before screening in ALIENTO, or within the 12 months before screening in ARNASA. †Patient-reported diagnosis of chronic bronchitis, defined as presence of cough or sputum on most days.								
Table 1: Baseline demographics and clinical characteristics (mITT population)								

hospitalisation for more than 24 h or death. Secondary efficacy endpoints at week 52 in both trials were annualised rate of severe exacerbations, annualised rate of moderate or severe exacerbations in participants with baseline blood eosinophil count of less than 300 cells/ μ L, time to first moderate or severe exacerbation, health-related quality of life per SGRQ-C total score (change from baseline and proportion of participants with clinically meaningful improvement [≥ 4 -point decrease from baseline]), and changes from baseline in the E-RS:COPD total score and in the post-bronchodilator FEV₁. In ALIENTO only, change from baseline in five-repetition sit-to-stand test time at week 52 was a secondary endpoint. All secondary outcomes are included in this report. For more information on the exploratory biomarker analyses and details of the ALIENTO interim analysis see appendix 1 (p 18). Safety was evaluated by adverse events, including infections and major adverse cardiovascular events (appendix 1 p 16), laboratory tests, vital signs, and electrocardiograms. Adverse event severity was graded according to the Division of AIDS Table version 2.1.

Statistical analysis

For each trial, assuming an annualised exacerbation rate of 1.5 in the placebo group, a 15% dropout rate, a two-sided significance level of 0.05, and a Poisson overdispersion parameter of 1.3, the planned sample size of 1290 participants provided about 90% power to detect a 20% reduction in annualised exacerbation rate

between each astegolimab group and the placebo group. Safety was monitored by an independent data monitoring committee.

Efficacy was evaluated in the modified intention-to-treat (mITT) population, which comprised participants who received one or more doses of the trial drug, based on randomised treatment assignment. Safety was evaluated in the safety evaluable population, which comprised participants who received one or more doses of the trial drug, based on treatment received. Hypothesis testing and estimation of treatment effects was done with regression models. The primary efficacy endpoint was evaluated using a negative binomial regression model, including the total number of events observed over the 52-week treatment period as the response variable, the treatment group and the stratification factors (smoking status at screening and region) as covariates in both trials, the number of moderate or severe COPD exacerbations within 12 months before screening as an additional covariate in ARNASA, and the natural log transformation of the participant's time at risk as an offset term. All missing data, including missing data due to deaths, were considered to be similar to data from other participants in the same treatment group with the same baseline characteristics and no such missing data (compatible with a missing at random assumption). Type I error was controlled using a fixed-sequence testing procedure at a 5% significance level for astegolimab every 2 weeks compared with placebo, then astegolimab every 4 weeks compared with placebo. Hypothesis testing

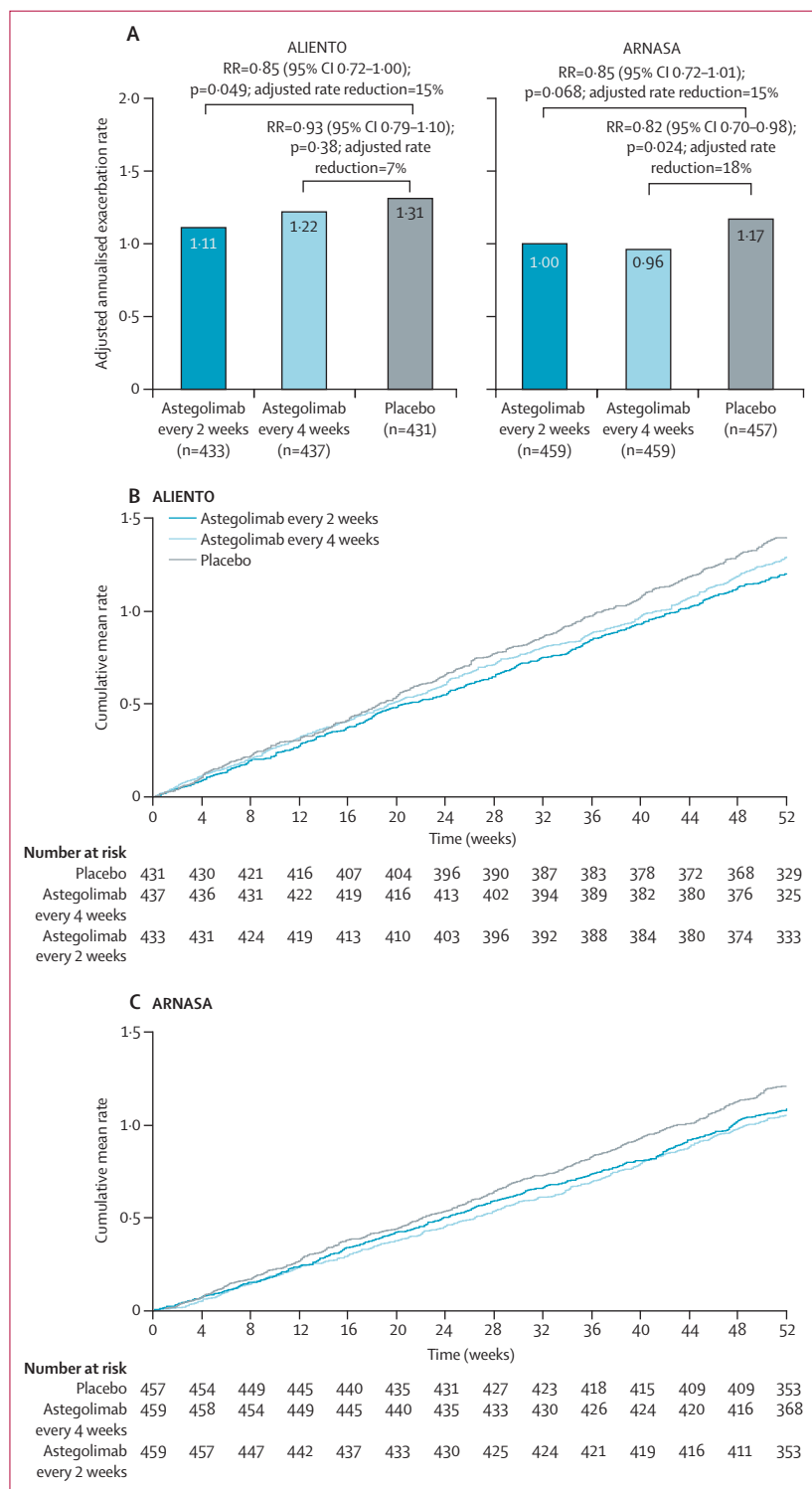


Figure 1: Primary endpoint in ALIENTO and ARNASA

(A) Adjusted annualised rate of moderate or severe COPD exacerbations. (B) Cumulative mean rate of moderate or severe COPD exacerbations for the mITT population in ALIENTO. (C) Cumulative mean rate of moderate or severe COPD exacerbations for the mITT population in ARNASA. COPD=chronic obstructive pulmonary disease. mITT=modified intention-to-treat. RR=rate ratio.

of secondary efficacy endpoints was gated on the success of the primary efficacy endpoint, and type I error was controlled within each treatment group using fixed-sequence testing procedures at a 5% significance level (appendix 1 p 26). Further details of the statistical analysis, including primary endpoint subgroup analyses and secondary endpoints, are in appendix 1 (pp 17–18). Safety results presented here are summarised descriptively.

Role of the funding source

The funders of the study contributed to the study design, data collection, data analysis, data interpretation, and writing of the report.

Results

In ALIENTO, 1722 participants were screened, 560 of whom did not meet the screening criteria; 140 of those participants were re-screened and subsequently enrolled. Between Oct 5, 2021, and Feb 19, 2024, 1302 participants were randomly assigned to astegolimab every 2 weeks (n=433), astegolimab every 4 weeks (n=438), or placebo (n=431), and included in the intention-to-treat (ITT) population (appendix 1 pp 32–33). In ARNASA, 2150 participants were screened, 934 of whom did not meet the screening criteria; 180 of those participants were re-screened and subsequently enrolled. Between Jan 9, 2023, and June 25, 2024, 1380 participants were randomly assigned to astegolimab every 2 weeks (n=460), astegolimab every 4 weeks (n=461), or placebo (n=459), and included in the ITT population (appendix 1 pp 34–35). One participant in ALIENTO and five participants in ARNASA did not receive treatment and were not included in the mITT or safety evaluable populations. Trial discontinuations over the 52-week treatment period were 16% in ALIENTO and 12% in ARNASA (appendix 1 pp 32–35).

Participant characteristics were generally balanced between treatment groups, and were generally similar across trials, with both trials having more than 85% of participants with blood eosinophil count less than 300 cells/ μ L, about 50% with blood eosinophil count less than 150 cells/ μ L, about 33% current smokers, and about 75% receiving triple inhaled maintenance therapy (table 1).

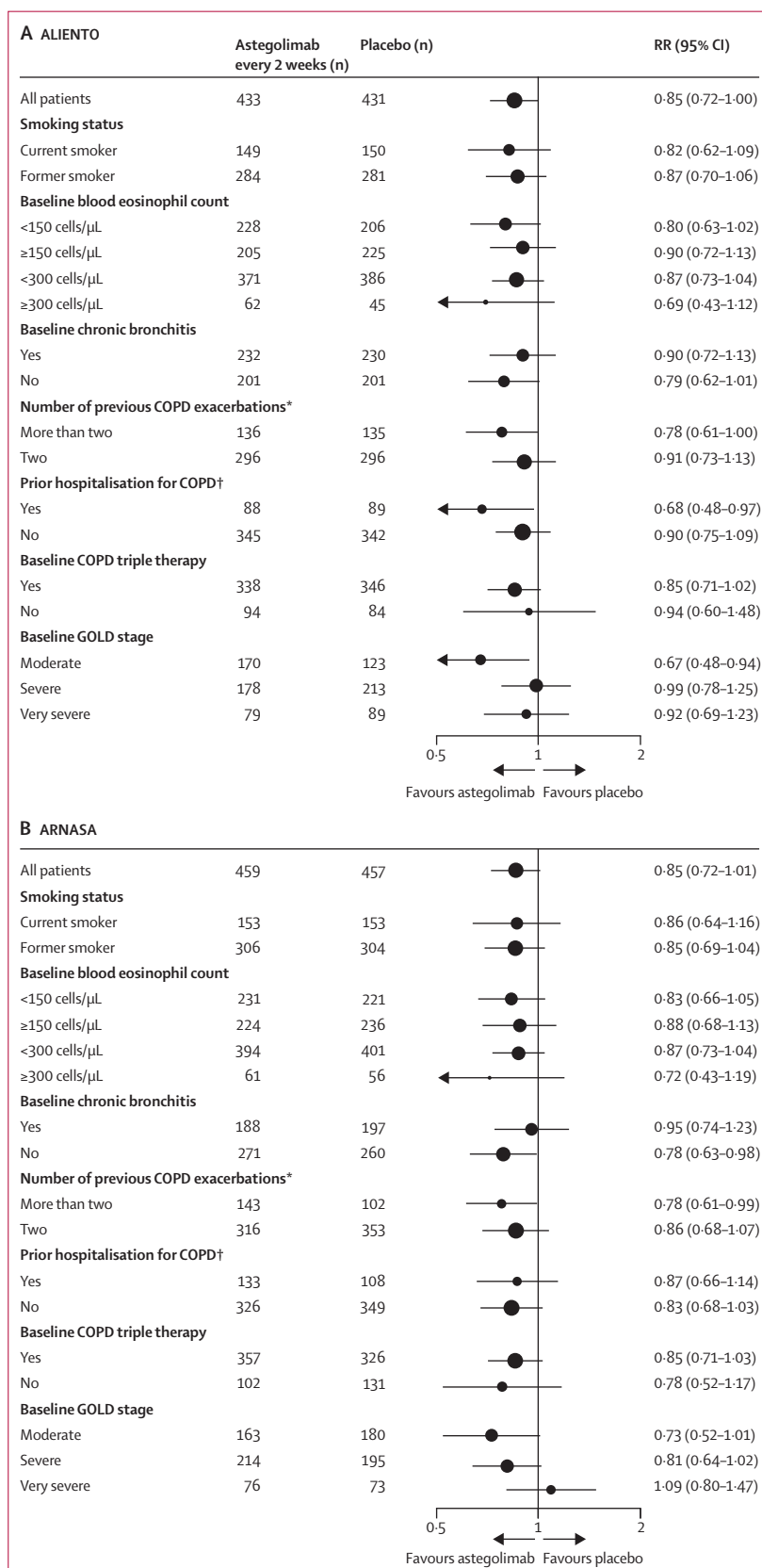
In ALIENTO, the unadjusted annualised rates of moderate or severe exacerbations were 1.21 for astegolimab every 2 weeks, 1.30 for astegolimab every 4 weeks, and 1.40 for placebo. The adjusted annualised rates of moderate or severe exacerbations were 1.11 for astegolimab every 2 weeks, 1.22 for astegolimab every 4 weeks, and 1.31 for placebo; adjusted rate ratios (RRs) versus placebo were 0.85 (95% CI 0.72–1.00; p=0.049) for astegolimab every 2 weeks and 0.93 (0.79–1.10; p=0.38; this result broke the ALIENTO fixed sequence testing procedure; appendix 1 p 26) for astegolimab every 4 weeks (figure 1A). In ARNASA, the unadjusted

annualised rates of moderate or severe exacerbations were 1·10 for astegolimab every 2 weeks, 1·06 for astegolimab every 4 weeks, and 1·22 for placebo. The adjusted annualised rates of moderate or severe exacerbations were 1·00 for astegolimab every 2 weeks, 0·96 for astegolimab every 4 weeks, and 1·17 for placebo; adjusted RRs versus placebo were 0·85 (95% CI 0·72–1·01; $p=0·068$; this result broke the ARNASA fixed sequence testing procedure; appendix 1 p 26) for astegolimab every 2 weeks and 0·82 (0·70–0·98; $p=0·024$) for astegolimab every 4 weeks (figure 1A). Cumulative mean rates of moderate or severe exacerbations for astegolimab every 2 weeks and every 4 weeks compared with placebo are provided for ALIENTO and ARNASA (figure 1B, C). For participants with more than two previous exacerbations (within a 12-month period in the 24 months before screening in ALIENTO, and in the 12 months before screening in ARNASA), the RRs for astegolimab every 2 weeks versus placebo were 0·78 (95% CI 0·61–1·00) in ALIENTO and 0·78 (0·61–0·99) in ARNASA. For participants with moderate airflow obstruction (GOLD 2), the RRs for astegolimab every 2 weeks versus placebo were 0·67 (0·48–0·94) in ALIENTO, and 0·73 (0·52–1·01) in ARNASA (figure 2). Other prespecified subgroup analyses of the primary endpoint, including by current or former smoking status and blood eosinophil count, showed a generally consistent treatment effect in favour of astegolimab every 2 weeks or every 4 weeks versus placebo across different subgroups. However, in these subgroup analyses, the 95% CIs crossed one and the observed effect sizes were small (figure 2; appendix 1 pp 36–41). Post-hoc analyses of the proportion of moderate or severe exacerbations in both trials that required treatment with antibiotics only, steroids only, or both are reported in appendix 1 (p 41).

Adjusted RRs for annualised rates of severe exacerbations (hospitalisation >24 hours or death) for astegolimab versus placebo were 0·71 (95% CI 0·49–1·05; $p=0·088$) for astegolimab every 2 weeks and 0·78 (0·53–1·14; $p=0·20$) for astegolimab every 4 weeks in ALIENTO, and 0·67 (0·47–0·95; $p=0·024$) for astegolimab every 2 weeks and 0·83 (0·59–1·16; $p=0·27$) for astegolimab every 4 weeks in ARNASA (table 2; appendix 1 p 42). Other exacerbation endpoints are shown in table 2.

Figure 2: Adjusted annualised rate of moderate or severe COPD exacerbations for astegolimab every 2 weeks versus placebo in ALIENTO and ARNASA, by selected subgroups (mITT population)

(A) ALIENTO. (B) ARNASA. COPD=chronic obstructive pulmonary disease. GOLD=Global Initiative for Chronic Obstructive Lung Disease. mITT=modified intention-to-treat. RR=rate ratio. *Number of exacerbations in a 12-month period in the 24 months before screening in ALIENTO or in the 12 months before screening in ARNASA. †Hospitalisation for COPD for more than 24 h within the last 12 months before screening.



	ALIENTO			ARNASA		
	Astegolimab every 2 weeks (n=433)	Astegolimab every 4 weeks (n=437)	Placebo (n=431)	Astegolimab every 2 weeks (n=459)	Astegolimab every 4 weeks (n=459)	Placebo (n=457)
Annualised rate of severe COPD exacerbations over the 52-week treatment period						
Adjusted annualised rate	0.18	0.20	0.26	0.14	0.17	0.20
Adjusted rate ratio vs placebo (95% CI)	0.71 (0.49 to 1.05); p=0.088	0.78 (0.53 to 1.14); p=0.20	..	0.67 (0.47 to 0.95); p=0.024	0.83 (0.59 to 1.16); p=0.27	..
Annualised rate of moderate or severe COPD exacerbations over the 52-week treatment period in participants with a baseline blood eosinophil count <300 cells/μL						
Adjusted annualised rate	1.12; n=371	1.20; n=386	1.29; n=386	0.99; n=394	0.97; n=392	1.13; n=401
Adjusted rate ratio vs placebo (95% CI)	0.87 (0.73 to 1.04); p=0.12	0.93 (0.78 to 1.11); p=0.42	..	0.87 (0.73 to 1.04); p=0.13	0.85 (0.72 to 1.02); p=0.085	..
Time to first moderate or severe COPD exacerbation during the 52-week treatment period						
Median time to first exacerbation (95% CI), weeks	42.4 (30.14 to NE)	39.9 (31.14 to 46.14)	33.7 (27.43 to 44.29)	47.1 (39.6 to NE)	45.6 (39.3 to NE)	38.7 (33.9 to 47.7)
Stratified hazard ratio vs placebo (95% CI)	0.88 (0.74 to 1.06); p=0.18	0.94 (0.79 to 1.13); p=0.53	..	0.90 (0.75 to 1.08); p=0.25	0.90 (0.75 to 1.07); p=0.23	..
Absolute change from baseline in SGRQ-C total score at week 52						
Adjusted mean (SE)	-6.75 (0.77); n=342	-6.25 (0.76); n=351	-4.87 (0.77); n=341	-7.37 (0.89); n=392	-7.64 (0.88); n=409	-6.60 (0.88); n=400
Difference in adjusted means vs placebo (95% CI)	-1.87 (-4.01 to 0.26); p=0.086	-1.37 (-3.50 to 0.75); p=0.21	..	-0.77 (-3.23 to 1.68); p=0.54	-1.04 (-3.48 to 1.40); p=0.40	..
Proportion of participants with a decrease from baseline of ≥ 4 points in SGRQ-C total score at week 52						
Adjusted responder rate	54.3%; n=413	53.9%; n=423	49.8%; n=406	58.5%; n=439	53.7%; n=449	48.5%; n=439
Odds ratio vs placebo (95% CI)	1.20 (0.90 to 1.60); p=0.21	1.18 (0.89 to 1.57); p=0.26	..	1.49 (1.14 to 1.96); p=0.0037	1.23 (0.94 to 1.61); p=0.13	..
Absolute change from baseline in E-RS:COPD total score at week 52						
Adjusted mean (SE)	-1.26 (0.30); n=326	-1.32 (0.30); n=328	-1.12 (0.30); n=327	-2.28 (0.28); n=359	-1.88 (0.28); n=383	-2.23 (0.28); n=370
Difference in adjusted means vs placebo (95% CI)	-0.14 (-0.97 to 0.69); p=0.74	-0.20 (-1.03 to 0.63); p=0.64	..	-0.05 (-0.83 to 0.73); p=0.90	0.35 (-0.42 to 1.12); p=0.37	..
Absolute change from baseline in post-bronchodilator FEV₁ at week 52						
Adjusted mean (SE), mL	-3 (12); n=316	-3 (11); n=333	-19 (12); n=317	-8 (11); n=390	-25 (11); n=394	-22 (11); n=393
Difference in adjusted means vs placebo (95% CI), mL	16 (-17 to 48); p=0.34	16 (-16 to 48); p=0.32	..	14 (-17 to 46); p=0.37	-3 (-34 to 29); p=0.87	..
Absolute change from baseline in five-repetition sit-to-stand test time at week 52						
Adjusted mean (SE), seconds	-1.38 (1.41); n=352	0.37 (1.41); n=349	-1.08 (1.42); n=348	NA	NA	NA
Difference in adjusted mean vs placebo (95% CI), seconds	-0.30 (-4.23 to 3.62); p=0.88	1.45 (-2.49 to 5.38); p=0.47	..	NA	NA	NA

For the hierarchy of testing for ALIENTO and ARNASA see appendix 1 (p 26). COPD=chronic obstructive pulmonary disease. E-RS:COPD=Evaluating Respiratory Symptoms in COPD. FEV₁=forced expiratory volume in 1 s. mITT=modified intention-to-treat. NA=not applicable. NE=not estimable. SGRQ-C=St George's Respiratory Questionnaire for COPD.

Table 2: Secondary efficacy endpoints (mITT population)

Adjusted mean treatment differences versus placebo for change from baseline in SGRQ-C total score at week 52 in ALIENTO were -1.87 (95% CI -4.01 to 0.26; p=0.086) and -1.37 (-3.50 to 0.75; p=0.21) for astegolimab every 2 weeks and every 4 weeks, respectively, and in ARNASA were -0.77 (-3.23 to 1.68; p=0.54) and -1.04 (-3.48 to 1.40; p=0.40), respectively. The odds ratios for proportion of participants with a ≥ 4 -point decrease from baseline in SGRQ-C total score versus placebo at week 52 were 1.20 (0.90 to 1.60; p=0.21) for astegolimab every 2 weeks and 1.18 (0.89 to 1.57; p=0.26) for astegolimab every 4 weeks in ALIENTO, and 1.49 (1.14 to 1.96; p=0.0037) for astegolimab every

2 weeks and 1.23 (0.94 to 1.61; p=0.13) for astegolimab every 4 weeks in ARNASA (table 2). No differences in change from baseline in E-RS:COPD score, post-bronchodilator FEV₁ or the five-repetition sit-to-stand test (ALIENTO only) were observed for either astegolimab group versus placebo (table 2). In both trials, target engagement was shown via soluble ST2 and blood eosinophil count changes (appendix 1 pp 43–44).

In both trials, the majority of participants in each group had one or more adverse event, with overall incidence occurring in 1093 (84.0%) of 1301 participants in ALIENTO and 1176 (85.5%) of 1375 in ARNASA (table 3). The majority of adverse events were grade 1 or 2 in

	ALIENTO				ARNASA			
	Astegolimab every 2 weeks (n=438)	Astegolimab every 4 weeks (n=435)	Placebo (n=428)	Total (n=1301)	Astegolimab every 2 weeks (n=464)	Astegolimab every 4 weeks (n=462)	Placebo (n=449)	Total (N=1375)
Participants with adverse events								
One or more adverse events	357 (81.5%)	363 (83.4%)	373 (87.1%)	1093 (84.0%)	386 (83.2%)	406 (87.9%)	384 (85.5%)	1176 (85.5%)
One or more serious adverse events	107 (24.4%)	122 (28.0%)	135 (31.5%)	364 (28.0%)	113 (24.4%)	124 (26.8%)	107 (23.8%)	344 (25.0%)
One or more adverse events of grade ≥ 3	147 (33.6%)	161 (37.0%)	175 (40.9%)	483 (37.1%)	135 (29.1%)	167 (36.1%)	122 (27.2%)	424 (30.8%)
One or more adverse events leading to withdrawal from trial	28 (6.4%)	23 (5.3%)	27 (6.3%)	78 (6.0%)	18 (3.9%)	21 (4.5%)	20 (4.5%)	59 (4.3%)
One or more adverse events leading to death	13 (3.0%)	11 (2.5%)	16 (3.7%)	40 (3.1%)	11 (2.4%)	18 (3.9%)	15 (3.3%)	44 (3.2%)
One or more adverse events of infection	205 (46.8%)	189 (43.4%)	182 (42.5%)	576 (44.3%)	227 (48.9%)	234 (50.6%)	216 (48.1%)	677 (49.2%)
One or more major adverse cardiovascular event*	17 (3.9%)	12 (2.8%)	17 (4.0%)	46 (3.5%)	8 (1.7%)	13 (2.8%)	13 (2.9%)	34 (2.5%)
Most common adverse events (occurring in $\geq 5\%$ of participants in any treatment group in either trial)								
COPD†	238 (54.3%)	254 (58.4%)	252 (58.9%)	744 (57.2%)	211 (45.5%)	231 (50.0%)	230 (51.2%)	672 (48.9%)
Upper respiratory tract infection	21 (4.8%)	19 (4.4%)	12 (2.8%)	52 (4.0%)	51 (11.0%)	50 (10.8%)	45 (10.0%)	146 (10.6%)
Nasopharyngitis	52 (11.9%)	46 (10.6%)	36 (8.4%)	134 (10.3%)	33 (7.1%)	43 (9.3%)	29 (6.5%)	105 (7.6%)
COVID-19	31 (7.1%)	43 (9.9%)	36 (8.4%)	110 (8.5%)	19 (4.1%)	24 (5.2%)	18 (4.0%)	61 (4.4%)
Pneumonia	29 (6.6%)	22 (5.1%)	22 (5.1%)	73 (5.6%)	30 (6.5%)	28 (6.1%)	35 (7.8%)	93 (6.8%)
Injection site reaction	20 (4.6%)	29 (6.7%)	26 (6.1%)	75 (5.8%)	17 (3.7%)	22 (4.8%)	17 (3.8%)	56 (4.1%)
Hypertension	20 (4.6%)	24 (5.5%)	24 (5.6%)	68 (5.2%)	20 (4.3%)	21 (4.5%)	21 (4.7%)	62 (4.5%)
Back pain	16 (3.7%)	27 (6.2%)	21 (4.9%)	64 (4.9%)	19 (4.1%)	20 (4.3%)	14 (3.1%)	53 (3.9%)
Dyspnoea	12 (2.7%)	12 (2.8%)	19 (4.4%)	43 (3.3%)	18 (3.9%)	20 (4.3%)	24 (5.3%)	62 (4.5%)
Diarrhoea	15 (3.4%)	18 (4.1%)	15 (3.5%)	48 (3.7%)	18 (3.9%)	14 (3.0%)	28 (6.2%)	60 (4.4%)
Urinary tract infection	18 (4.1%)	15 (3.4%)	22 (5.1%)	55 (4.2%)	12 (2.6%)	18 (3.9%)	13 (2.9%)	43 (3.1%)
Headache	13 (3.0%)	16 (3.7%)	18 (4.2%)	47 (3.6%)	12 (2.6%)	18 (3.9%)	23 (5.1%)	53 (3.9%)

Data are n (%). COPD=chronic obstructive pulmonary disease. *Recorded as adverse events of special interest; major adverse cardiovascular events were defined as death due to cardiovascular causes, non-fatal myocardial infarction, non-fatal stroke or transient ischaemic attack, unstable angina, cardiac chest pain requiring hospitalisation, coronary revascularisation, and congestive heart failure requiring hospitalisation. †Medical occurrences or symptoms of COPD deterioration that were anticipated as part of the disease were not recorded as adverse events; however, deterioration that was judged by the investigator to have unexpectedly worsened in severity or frequency or changed in nature at any time during the trial was recorded as an adverse event, and all COPD exacerbations were recorded as adverse events or serious adverse events.

Table 3: Overview of adverse events (safety evaluable population)

intensity in both trials. Grade 3 or higher events occurred in 147 (33.6%) of 438 participants receiving astegolimab every 2 weeks, 161 (37.0%) of 435 participants receiving astegolimab every 4 weeks, and 175 (40.9%) of 428 participants receiving placebo, in ALIENTO, and in 135 (29.1%) of 464 participants receiving astegolimab every 2 weeks, 167 (36.1%) of 462 participants receiving astegolimab every 4 weeks, and 122 (27.2%) of 449 participants receiving placebo, in ARNASA (table 3). Deaths occurred in 13 (3.0%) of 438 participants receiving astegolimab every 2 weeks, in 11 (2.5%) of 435 receiving astegolimab every 4 weeks, and in 16 (3.7%) of 428 receiving placebo in ALIENTO, and in 11 (2.4%) of 464, 18 (3.9%) of 462, and 15 (3.3%) of 449, respectively, in ARNASA. In ALIENTO, there was one death (0.2%) from ischaemic stroke in the astegolimab every 2 weeks group that was considered to be related to treatment by the investigator. In ARNASA, there were two deaths (0.4%) in the astegolimab every two weeks group that were considered to be related to treatment by the investigator; one patient died due to COVID-19 and the other due to COPD. There were no meaningful

differences in the incidence of infections or major adverse cardiovascular events between astegolimab and placebo groups, nor any consistent trends in changes in laboratory tests, vital signs, and electrocardiograms across both trials.

Discussion

ALIENTO and ARNASA are the first pivotal trials reporting data for targeting the ST2/IL-33 pathway in COPD. Although a significant reduction in the annualised rate of moderate or severe exacerbations was only observed in ALIENTO, the primary endpoint results from both trials were numerically consistent with a 15% reduction seen in the every 2 weeks group versus the placebo group of both trials. Furthermore, a consistent reduction in the annualised rate of severe exacerbations with astegolimab every 2 weeks versus placebo was observed in both ALIENTO (29%) and ARNASA (33%). As testing hierarchy was broken early for both studies, statistical significance can only be claimed for the every 2 weeks primary endpoint in ALIENTO. For astegolimab every 4 weeks, there was a non-statistically significant

reduction in annualised rate of moderate or severe exacerbations of 7% in ALIENTO and 18% in ARNASA (non-significant due to a break in testing hierarchy). Since the overall benefit–risk profile across both trials favoured the astegolimab every 2 weeks regimen, this is the focus of the discussion.

COPD is most commonly associated with neutrophilic inflammation or low blood eosinophil counts (<300 cells/ μ L).¹¹ These findings from ALIENTO and ARNASA were observed in participants who were enrolled regardless of evidence of eosinophilic inflammation, with more than 85% of participants having baseline blood eosinophil count of less than 300 cells/ μ L and about 50% with eosinophils less than 150 cells/ μ L (mean 181–184 cells/ μ L) across trials. Therefore, although we observed a somewhat modest, statistically significant 15% reduction in moderate or severe exacerbations with astegolimab every 2 weeks in ALIENTO, this was in individuals with a predominantly non-eosinophilic phenotype who continue to exacerbate despite optimised inhaled therapy. This phenotype is less modifiable by corticosteroids than the eosinophilic phenotype²² and therefore the observed reduction in exacerbations is clinically meaningful. By comparison, advances in COPD therapy from dual-inhaled therapy to triple-inhaled therapy reduced the rate of moderate or severe exacerbations by about 13–25%.^{23–25} The 15% reduction in exacerbation rate in ALIENTO and ARNASA is therefore of similar magnitude to that observed in previous treatment advances.

Furthermore, subgroup analysis of ALIENTO and ARNASA showed that reductions in exacerbation frequency were not driven by key subgroups defined by blood eosinophil count or current or former smoking status, suggesting that astegolimab has potential efficacy across a clinically heterogeneous patient population, and might offer an additional treatment option for patients who are not eligible for currently approved add-on COPD biologics. For example, other phase 3 trials of COPD biologics were conducted in more restricted populations than in ALIENTO and ARNASA, including those of dupilumab (requiring an eosinophilic phenotype and chronic bronchitis),^{8,9} and mepolizumab (requiring an eosinophilic phenotype).¹⁰ The ALIENTO and ARNASA trials have provided preliminary information on the potential treatment benefits of astegolimab in other specific subgroups of participants, including potentially greater treatment benefits in those with a recent history of more than two exacerbations, those with GOLD 2 (moderate) airway obstruction, and those without a diagnosis of chronic bronchitis (defined as patient-reported absence of cough or sputum on most days); however, additional responder analyses will be essential to identify the patients with COPD who would benefit most from astegolimab in future.

Compared with placebo, astegolimab every 2 weeks treatment was also associated with non-significant, but

consistent, reductions in the annualised rates of severe exacerbations (ie, those leading to hospitalisation for more than 24 h or death) of 29% in ALIENTO and 33% in ARNASA (non-significant due to the early break in the ARNASA testing hierarchy), respectively. Compared with moderate exacerbations, severe exacerbations are associated with an elevated risk of subsequent exacerbations, cardiovascular events, hospitalisations and death,^{3,6} and have a greater negative impact on quality of life and activity.²⁶ Annualised severe exacerbation rates in the placebo groups for ALIENTO and ARNASA were higher than seen in recent pivotal trials of other biologics^{8,10} and, coupled with worse outcomes in patients with low blood eosinophils (vs high eosinophils) following hospitalisation with a COPD exacerbation,²⁷ a reduction in severe exacerbations would be of clinical benefit. Furthermore, hospitalisations are the biggest contributor to COPD health-care costs in Europe and Canada,^{28,29} and COPD-related costs are predicted to rise to US\$60·5 billion in the USA alone by 2029.³⁰ Overall, the findings support further work to understand the clinical and economic impact of astegolimab, with respect to the potential for astegolimab to reduce severe exacerbations.

Changes in soluble ST2 and blood eosinophil counts showed target engagement, consistent with previous trials of astegolimab.^{19,20,31} Both astegolimab doses were well tolerated and the safety profile was consistent with previous reports.^{19,20,32–34} Based on the role of ST2 in the inflammatory response and some preclinical models in cardiomyocytes, there is a theoretical risk of increased susceptibility to infection and major adverse cardiovascular events when targeting the ST2/IL-33 pathway,^{35,36} and these trials are the first large-scale trials to generate clinical major adverse cardiovascular events data following ST2 inhibition with astegolimab in COPD. However, aligned with recent reports,³² no increased infection or major adverse cardiovascular events risk was observed during ALIENTO or ARNASA.

A strength of ALIENTO and ARNASA lies in the large, representative, international nature of the trials, with a combined diverse population of 2676 participants recruited from 39 countries, including about 21% of participants from racial groups other than White. Hispanic participants were well represented in the trials, at nearly 19% of participants in ALIENTO and 15% in ARNASA. Black or African American participants made up 2·7% of participants in ALIENTO and 5·5% in ARNASA. The age range of participants was 41–90 years in ALIENTO and 40–81 years in ARNASA, with 64% and 66%, respectively, being 65 years or older, reflective of COPD being most prevalent in older age groups.^{37,38} Both trials enrolled more men than women with ALIENTO having 56% versus 44% and ARNASA having 64% versus 36%, the latter of which is in line with the 2:1 ratio observed globally.³⁷ Furthermore, participants were recruited regardless of blood eosinophil count, current or

former smoking status, or chronic bronchitis diagnosis. This approach was informed by early phase 2 trials of anti-ST2/IL-33 biologics, which did not identify a consistent responder subgroup,^{19,39,40} most likely reflecting the heterogeneity of the underlying biology, as the ST2 receptor is expressed on a wide range of immune and non-immune cells, and IL-33 is released in response to several exacerbation triggers. As such, the broad population in ALIENTO and ARNASA allows for examination of a range of clinical and biological characteristics to potentially identify responder subgroups.

Limitations include the observed annualised exacerbation rates in the placebo groups being lower than the assumed rate of 1.5, which reduced the trial powering. Although broader than some previous trials of COPD biologics,^{8–10} the selective eligibility criteria (as is typical of randomised clinical trials) mean that the trial population might not be fully reflective of that seen in wider clinical practice; for example, in relation to the presence or severity of comorbidities, which are common in patients with COPD.⁴¹

In conclusion, astegolimab at a dose of 476 mg every 2 weeks was associated with a lower annual rate of moderate or severe exacerbations versus placebo in patients with COPD and a history of frequent exacerbations in ALIENTO. The effect size was consistent in both studies but did not meet statistical significance in ARNASA. These findings suggest that targeting the ST2/IL-33 pathway might have a therapeutic role in patients with COPD with high symptom and clinical burden, who have limited treatment options available to them.

Contributors

AP was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. NJG made substantial contributions to the conception or design of the work, and was involved in data interpretation and drafting and reviewing of the manuscript. SPB was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. NR was involved in data interpretation and drafting and reviewing of the manuscript. BC was involved in data interpretation and drafting and reviewing of the manuscript. JAW was involved in data interpretation and drafting and reviewing of the manuscript. AA was involved in data interpretation and drafting and reviewing of the manuscript. RT-S made substantial contributions to the conception or design of the work, and was involved in data interpretation and drafting and reviewing of the manuscript. MKH made substantial contributions to the conception or design of the work, and was involved in data interpretation and drafting and reviewing of the manuscript. WJ was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. NAH was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. PN made substantial contributions to the conception or design of the work, and was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. PB was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. KP was involved in data acquisition and drafting and reviewing of the manuscript. YS was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. SK was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. TY was involved in data acquisition, data interpretation, and drafting and reviewing of the manuscript. OG was

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Declaration on interests

AP has received grants or contracts from AstraZeneca, Chiesi, GlaxoSmithKline, and Sanofi; consulting fees from AstraZeneca, Avillion, Chiesi, GlaxoSmithKline, Moderna, Regeneron, F Hoffmann-La Roche, Sanofi, Zambon, and Zentiva; and honoraria from AstraZeneca, Avillion, Chiesi, Regeneron, Moderna, GlaxoSmithKline, F Hoffmann-La Roche, Sanofi, and Zambon. NJG has received support for this study from the National Institute for Health and Care Research; grants or contracts from GlaxoSmithKline, AstraZeneca, and Genentech; consulting fees from AstraZeneca, Genentech, GlaxoSmithKline, Pulmonx, F Hoffmann-La Roche, and Sanofi; honoraria from AstraZeneca, Chiesi, Genentech, GlaxoSmithKline, Pulmonx, F Hoffmann-La Roche, and Sanofi; and support for travel from AstraZeneca, GlaxoSmithKline, Sanofi, and Chiesi. SPB has received grants or contracts from the National Institutes of Health, NuVaira, Sanofi, Regeneron, Genentech, COPD Foundation, Apreo, Uniquity, and Connect Biopharma; royalties from Springer; consulting fees from Sanofi, Regeneron, Boehringer Ingelheim, Apreo, Genentech, Chiesi, AstraZeneca, GlaxoSmithKline, Kymera, Merck & Co, Verona Pharma, Polarean, Uniquity, and Connect Biopharma; honoraria from IntegrityCE, Medscape, Horizon CME, Illuminate Health, and Integrity Communications; and is chair of the Program Committee of the Clinical Problems Assembly of the American Thoracic Society. NR has received grants or contracts from Chiesi, GlaxoSmithKline, and Pfizer; consulting fees from GlaxoSmithKline, AstraZeneca, Sanofi, Chiesi, NuVaira, Pfizer, Austral, Biosency, and F Hoffmann-La Roche; honoraria from GlaxoSmithKline, AstraZeneca, Sanofi, Chiesi, Pfizer, Zambon, Merck & Co, and Menarini; and support for travel from Chiesi, AstraZeneca, and GlaxoSmithKline. BC has received grants or contracts from GlaxoSmithKline, AstraZeneca, Menarini, Sanofi, Axios, Chiesi, F Hoffmann-La Roche, and Genentech; consulting fees from GlaxoSmithKline, AstraZeneca, and Sanofi; honoraria from GlaxoSmithKline, AstraZeneca, Menarini, Chiesi, and Regeneron; support for travel from GlaxoSmithKline and Sanofi; and has participated in a data safety monitoring board or advisory board for AstraZeneca, Sanofi, and Vertex. JAW has received grants from AstraZeneca, Boehringer Ingelheim, Chiesi, GlaxoSmithKline, and Novartis; consulting fees from Altesa, AstraZeneca, CSL Behring, Epiendo, F Hoffman-La Roche, GlaxoSmithKline, Empirico, Gilead, MucPharm, NEATstix, Pfizer, and Sanofi; honoraria from Altesa, AstraZeneca, Boehringer Ingelheim, F Hoffman-La Roche, GlaxoSmithKline, Handok, Novartis, Recipharm, and Sanofi; and has

participated in a data safety monitoring board for Virtus. AA has received consultancy fees from GlaxoSmithKline, AstraZeneca, Chiesi, F Hoffmann-La Roche, and Menarini; honoraria from GlaxoSmithKline, AstraZeneca, Chiesi, F Hoffmann-La Roche, Menarini, Zambon, and Glenmark; support for attending meetings or travel from F Hoffmann-La Roche; and is chair of the board of directors of GOLD. RT-S has received grants or contracts from GlaxoSmithKline, F Hoffmann-La Roche, and Boehringer Ingelheim; consulting fees from GlaxoSmithKline, AstraZeneca, Johnson & Johnson, Renovion, F Hoffmann-La Roche, ENA Respiratory, Samay Health, Boehringer Ingelheim, COPD Foundation, Global Allergy and Airways Patient Platform, and Global Skin; honoraria from GlaxoSmithKline and F Hoffmann-La Roche; has or had a leadership or fiduciary role at COPD Foundation, ENA Respiratory, and Global Allergy and Airways Patient Platform; and owns share options at ENA Respiratory. MKH has received grants from the National Institutes of Health, Novartis, Sunovion, Nuvera, Sanofi, AstraZeneca, Boehringer Ingelheim, Galvanize Therapeutics, Biodesix, the COPD Foundation and the American Lung Association; honoraria from GlaxoSmithKline, AstraZeneca, Boehringer Ingelheim, Cipla, Chiesi, Novartis, Pulmonx, Teva, Verona, Merck & Co, Mylan, Sanofi, Genentech, F Hoffmann-La Roche, DevPro, Aerogen, Polarian, Regeneron, Amgen, UpToDate, Altesa Biopharma, CSL Behring, Bristol Myers Squibb, Zymeworks, Zambon, Windward Bio, Uniquity, Medscape, NACE, MDBriefcase, and Integrity; has participated in a data safety monitoring board for Novartis and Medtronic; and has stock options from Meissa Vaccines and Altesa Biopharma. WJ has received grants or contracts from AstraZeneca, Chiesi, and GlaxoSmithKline; consulting fees from AstraZeneca, Chiesi, F Hoffmann-La Roche, Sanofi, and GlaxoSmithKline; and honoraria from AstraZeneca, Chiesi, F Hoffmann-La Roche, Sanofi, and GlaxoSmithKline. NAH has received grants or contracts from GlaxoSmithKline, Sanofi, Genentech, Regeneron, Amgen, AstraZeneca, and Chiesi; consulting fees from GlaxoSmithKline, Sanofi, Genentech, Regeneron, AstraZeneca, Chiesi, Verona Pharma, and Connect Biopharma; and honoraria from Sanofi, Regeneron, AstraZeneca, and Genentech. PN has received grants or contracts from AstraZeneca, Teva, Sanofi, Foresee, Cyclomedica, F Hoffmann-La Roche, Genentech, Methapharm, and Eli Lilly; and personal fees from AstraZeneca, Sanofi, Arrowhead Pharma, GlaxoSmithKline, and Methapharm. PB has received grants or contracts from Boehringer Ingelheim; honoraria from Boehringer Ingelheim; support for travel from Boehringer Ingelheim; and is a board member of St John of God Health Care. KP has received grants or contracts from GlaxoSmithKline and AstraZeneca; honoraria from GlaxoSmithKline, AstraZeneca, Menarini, and Chiesi; and support for travel from GlaxoSmithKline, AstraZeneca, Menarini, Chiesi. YS has received honoraria from AstraZeneca, Merck & Co, Kracie, Sanofi, Boehringer Ingelheim, and GlaxoSmithKline. SK received grants or contracts from GlaxoSmithKline, AstraZeneca, Sanofi, Chiesi, F Hoffmann-La Roche, and Genentech; consulting fees from GlaxoSmithKline, AstraZeneca, Chiesi, and Sanofi; honoraria from GlaxoSmithKline, AstraZeneca, and Chiesi; and support for travel from GlaxoSmithKline and Sanofi. TY received grants from GlaxoSmithKline, AstraZeneca, Sanofi, F Hoffmann-La Roche, National Natural Science Foundation of Beijing, National Natural Science Foundation of China, and Ministry of Science and Technology of China. OG is an employee of F Hoffmann-La Roche and owns stocks or stock options in the company. RS is an employee of Genentech with stocks and stock options in F Hoffmann-La Roche. JN is an employee of Genentech with stocks in F Hoffmann-La Roche. DSC is an employee of Genentech, with stocks in F Hoffmann-La Roche and is a patent holder for method of treating COPD with an ST2 antagonist. JD is an employee of F Hoffmann-La Roche and owns stock in the company. MAG is an employee of Genentech with stocks and stock options in F Hoffmann-La Roche. WZ is an employee of Genentech with stocks in F Hoffmann-La Roche. XY is an employee of Genentech with stocks and stock options in F Hoffmann-La Roche–Genentech. DM is an employee of Genentech, with stocks in F Hoffmann-La Roche and is a patent holder for method of treating COPD with an ST2 antagonist. CFV has received grants or contracts from the German Ministry of Education and Science, AstraZeneca, Boehringer Ingelheim, Chiesi, CSL Behring, GlaxoSmithKline, and Grifols; consulting fees from Aerogen,

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Data sharing

Qualified researchers may request access to individual patient-level data through the clinical trial data request platform (<https://vivli.org/>). Further details on Roche's criteria for eligible studies are available on <https://vivli.org/members/ourmembers/>. For further details on Roche's Global Policy on the Sharing of Clinical Information and how to request access to related clinical trial documents, see <https://www.roche.com/innovation/process/clinical-trials/data-sharing>.

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